



Chronology and environmental changes from a sediment core spanning the last 485 ka from Lake Acıgöl (SW Anatolia)

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ABSTRACT

This study details the construction of an age-depth model for the uppermost 128 m of the 600 m long Acıgöl2009-B03 sediment core, retrieved from the hypersaline Lake Acıgöl in southwestern Anatolia. The model aligns the arboreal pollen record from Acıgöl2009-B03 with the LR04 benthic $\delta^{18}\text{O}$ stack marine oxygen isotope record. Initial correlation was achieved using the Dynamic Time Warping algorithm, which was subsequently refined through manual tuning and finalized with Bayesian age-depth modeling to enhance precision and report the associated uncertainty levels. The model's accuracy incorporates multiple chronological constraints, including three radiocarbon dates, three U/Th dates, and the Kos Plateau Tuff, dated at 161.3 ± 0.1 ka. According to this model, the upper 128 m of the Acıgöl2009-B03 sequence spans approximately the last 485,000 years, encompassing Marine Isotope Stages (MIS) 1–12 and part of MIS 13. Our model serves as an update to the previously published, linearly constructed, age model as being used more anchor points and an efficient algorithm for similarity measurements which lies on a robust statistical foundation. In this period, arboreal pollen data suggest increasing (decreasing) arboreal vegetation input during interglacial (glacial) periods.

1. Introduction

Long, continuous terrestrial paleoclimate records are essential for both regional and global paleoenvironmental reconstructions, despite their limited spatial distribution. These archives provide comprehensive evidence—including physical, chemical, and biological data—for reconstructing Quaternary environmental conditions and evaluating the effects of anthropogenic climate change. High-resolution and, temporally extensive datasets are particularly crucial for interpreting regional paleoclimatic changes. This is especially relevant in regions characterized by complex short- and long-term climate fluctuations, such as the eastern Mediterranean region. (e.g., Tzedakis, 2007; Ön and Özeren,

2019)

Southwest Anatolia, including the “Lake District” (Göller Bölgesi in Turkish), is distinguished for its numerous large high-altitude lakes such as Lake Salda, Lake Acıgöl, Lake Burdur, and Lake Eğirdir, among others. Previous climate studies in this region have primarily relied on analyses of short lake sediment cores (Tudryn et al., 2013; Kaiser et al., 2016; Balcı et al., 2018, 2020; Biltekin et al., 2018; Danladi and Akçer-Ön, 2018; Danladi et al., 2021, 2023; Doğan et al., 2024) or low-resolution assessments of Pleistocene tufa deposits (Tagliasacchi and Kayseri-Özer, 2020). Although a few temporally extensive Quaternary sedimentary archives exist within the broader region (Tzedakis et al., 2006; Badertscher et al., 2011; Litt and Anselmetti, 2014; Wagner et al., 2019)

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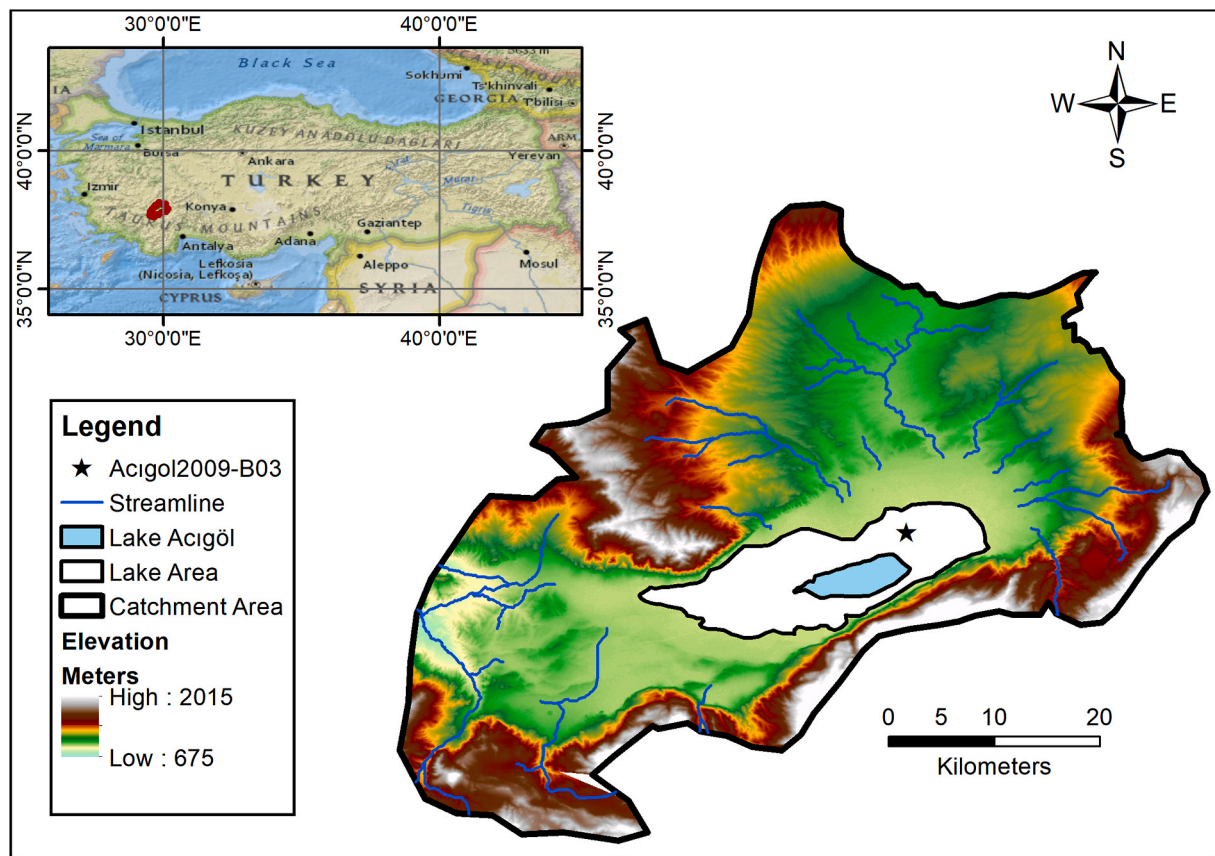


Fig. 1. Site location and watershed delineation map of Lake Acıgöl, shown with a digital elevation model (DEM). The map indicates the Acıgöl2009-B03 core location, streamlines, elevation contours, and the lake area. The Lake District map is shown in the top-right inset.

until recently, high-resolution records covering multiple glacial-interglacial cycles were unavailable for western Anatolia. However, it is well established that arboreal vegetation predominates during warm interglacial periods, whereas cooler phases are characterized by a decline in tree cover in the eastern Mediterranean region (e.g., Cheddadi and Rossignol-Strick, 1995; Rossignol-Strick and Paterne, 1999; Tzedakis et al., 2006).

Establishing precise chronological frameworks is essential for accurately interpreting these paleoenvironmental archives, requiring the alignment of temporal or spatial proxies through conventional correlation techniques. However, such approaches can yield misleading results without robust statistical frameworks (Pälike, 2001). Dynamic Time Warping (DTW), an algorithm based on dynamic programming, is utilized to assess similarities between sequences that exhibit temporal or spatial variations (Senin, 2008). Our assumption in this study is that arboreal pollen percentage is influenced by global climatic changes. This assumption is based on the observation that increases in arboreal pollen percentages are clearly visible during warm periods, even prior to the refinement of the age model applied in this study, which was initially based on a preliminary age model using dated levels. Moreover, arboreal pollen records, spanning multiple glacial and interglacial periods in the region, consistently show an increase during interglacial periods and a decrease during glacial periods (Tzedakis et al., 2006; Litt et al., 2014; Wagner et al., 2019). Based on this assumption, we applied DTW to correlate the arboreal pollen percentage from core Acıgöl2009-B03 with the global benthic stack marine oxygen isotope curve, thereby generating an age-depth model for the core. The observed similarities between the global isotope stack and pollen data are consistent with findings from previous studies (Lisiecki and Raymo, 2005; Railsback et al., 2015; Sadori et al., 2016).

In 2009, the Acıgöl2009-B03 sediment core was extracted, and initial

findings describing the geological, geochemical, and tectonosedimentary characteristics of Lake Acıgöl were published by Helvacı et al. (2013). Subsequently, Demory et al. (2020) constructed a linear age-depth model for the entire 600-m sedimentary sequence of Acıgöl2009-B03 using five paleomagnetic tie points. According to Demory et al. (2020), Acıgöl2009-B03 spans the last 2.3 million years. Acıgöl2009-B03 with the age model of Demory et al. (2020) was recently used by Andrieu-Ponel et al. (2021) to investigate the emergence of proto-cereals in the region and their connections with large herbivores and hominins in western Anatolia.

The primary aim of this study is to establish a chronological framework for the Acıgöl2009-B03 sediment core by leveraging the similarity between the arboreal pollen record from core Acıgöl2009-B03 and the global Marine Isotope Stages (MIS), which represent alternating warm and cool periods inferred from oxygen isotope data in deep-sea cores (Lisiecki and Raymo, 2005). The Acıgöl2009-B03 sediment core lacks pollen between depths of 128 and 153.7 m, thereby precluding the construction of a reliable chronology solely through the dynamic time warping (DTW) approach based on arboreal pollen percentages. Additionally, beyond a depth of 153.7 m, the core contains no datings, apart from the paleomagnetic transition dates reported in Demory et al. (2020). Therefore, to ensure a robust chronology, this study presents a timeline only for the upper 128 m of the core. Further refinements to the age model constructed by DTW algorithm involved incorporating 25 minor tuning points derived from seven primary age control points to mitigate artifacts introduced by the statistical model. Finally, a Bayesian age-depth model was constructed using the dated depths and tie points, ensuring a more robust and accurate chronology capable of adequately reflecting the associated uncertainties. Our results indicate that the 128-m sequence of core Acıgöl2009-B03 spans approximately the last 485,000 years, encompassing Marine Isotope Stages 1–12 and a

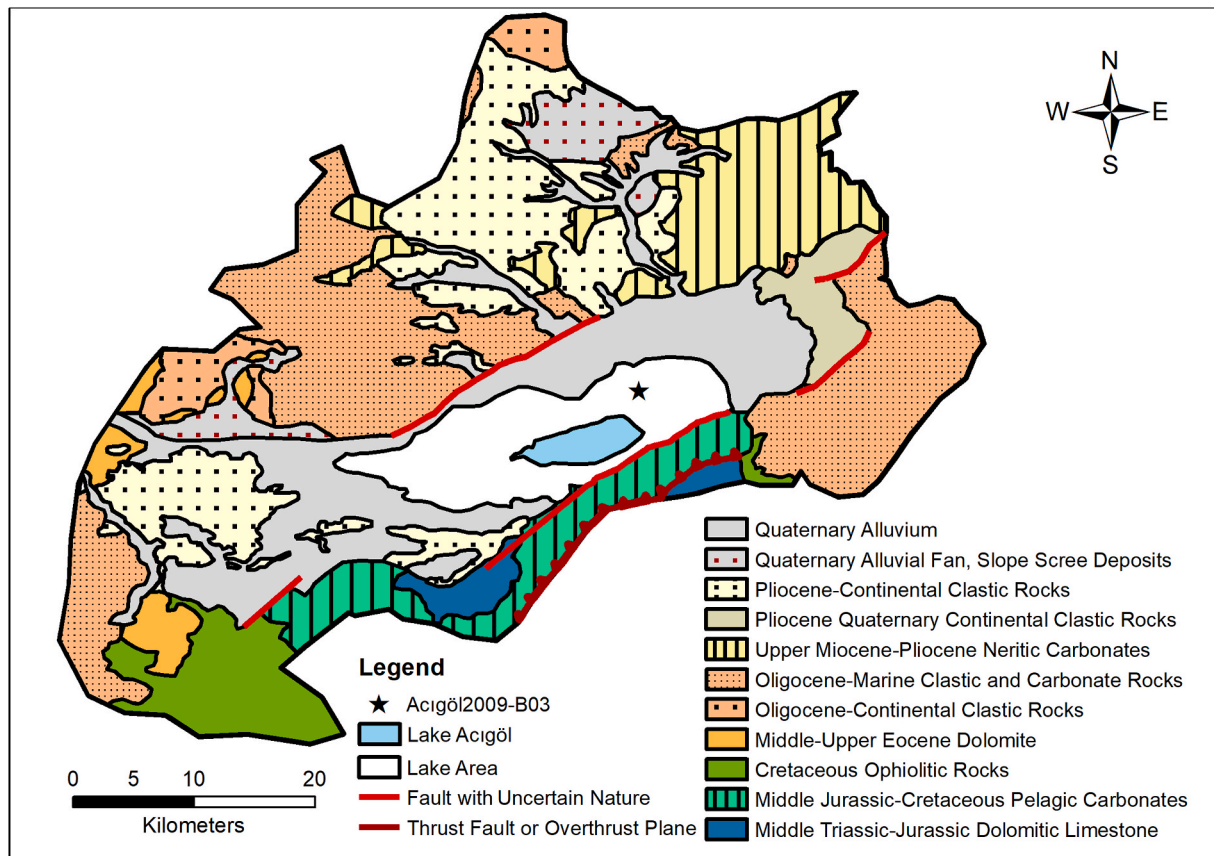


Fig. 2. Geological map of Lake Acıgöl and its surroundings. The map was revised from 1/100,000 geological map sections belong to M23 and M24 of General Directorate of Mineral Research and Exploration (MTA) (Konak et al., 2015).

significant portion of Stage 13. Our suggested model, constructed using DTW and refined with 25 tie points and 7 age control points, represents a significant improvement over the previously established age-depth model of Demory et al. (2020). The previous model was based on linear interpolation between the surface of the core and a paleomagnetic tie point at 178.5 m for the same 128-m chronostratigraphic interval.

The study aims to develop a robust age model for the longest core taken from Anatolia using statistical methods, and to enable future high-resolution paleoenvironmental and paleoclimate studies on this core to be conducted with more accurate age models. Environmental reconstructions based on accurate high-resolution age-depth model of the Acıgöl2009-B03 succession are crucial for the region, which constitutes an important route of early hominin migration between Africa and Eurasia, where the oldest hominin fossil (1.2–1.6 Myr; Lebatard et al., 2014) was discovered just 40 km west of Lake Acıgöl (Kappelman et al., 2007; Vialet et al., 2018). Therefore, we also briefly discuss the environmental evolution of the region lithology. We also believe that this sediment core, which was taken by Alkim Company to determine the production reserve in Acıgöl, where sodium sulfate and table salt are produced, will also provide important insights into understanding last 485,000 years of Anatolia and its surroundings.

2. Regional setting

The Acıgöl Basin is a hydrologically closed system undergoing progressive shrinkage primarily due to drought and agricultural water usage (Karaman et al., 2015). It is a hypersaline lake characterized by a NaCl (SO₄-Mg) rich brine composition slightly alkaline, with pH levels ranging between 7.1 and 8.8 (Mutlu et al., 1999). It features temporary dry mud flats and lacks perennial streams, relying instead mainly on small intermittent streams and groundwater inflow as its main water

sources (Özdemir and Bahadır, 2010; Balci et al., 2018).

Geomorphologically, the Acıgöl Basin is a NE-SW trending graben situated within the larger Burdur basin, straddling the Denizli-Afyon provincial border. The sub-basin spans approximately 45 km in length and 14 km in width, with the lake surface located at an elevation of 836 m above sea level. The Maymundağı (1689 m) and Yandağ (2033 m) mountains flank the basin to the northwest and southeast, respectively (Fig. 1). Lake Acıgöl currently covers an area of 450 ha and accumulates carbonates (dolomite and calcite), as well as minerals such as sodium and calcium sulfate (gypsum, mirabilite, glauberite), which are mined for soda production.

The geological features of the region have evolved under significant Neotectonic stresses, resulting in a highly diverse and complex structure (Şengör and Yılmaz, 1981). The basement of the region comprises Cretaceous pelagic carbonates, specifically the Yandağ limestone, and ophiolitic rocks like the Marmaris ophiolites. Surrounding the lake, the flat areas are covered by Oligocene marine clastic and carbonate rocks known as the Acıgöl Group. In the Pliocene epoch, continental clastic rocks of the Hayrettin Group dominate, comprising lacustrine sediments. The Cretaceous- Oligocene formations contribute to the elevated terrain north and south of the basin (Göktaş et al., 1989). A notable geological structure in the region is the Acıgöl Fault in the southeastern part of the graben, delineating the boundary between limestone and dolomitic limestones of the Lycian nappes. Conversely, the Maymundağı Fault in the northwest of the graben intersects Paleogene and Neogene units, though it is less visibly exposed compared to the Acıgöl Fault (Fig. 2, Bahadır and Özdemir, 2011).

Karstic forms are notably present, particularly in the southwestern part of Lake Acıgöl, where depressions (poljes) occur in several locations such as Ayvaz, Çaltı, and Çorakgöl. Along the slopes and foothills of Yandağ facing Acıgöl, numerous karstic formations oriented in a

Southwest-Northwest direction align with the Acıgöl Fault. Karst water outlets are observed in southwest of Acıgöl (Kuşcu et al., 2017).

The Acıgöl basin experiences a semiarid climate primarily due to the barrier effect of high-altitude mountain ranges surrounding the Anatolian Peninsula, which obstruct humid air masses. Current climate of the Acıgöl region is characterized by cold and rainy/snowy winters, hot and relatively dry summers. Annual precipitation in the region averages around 400 mm, while yearly evaporation in the lake is about 750 mm (based on averages from the Burdur weather station, see Fig. S1). Evidence of lacustrine terraces, reaching heights of up to 34 m above the current lake level, suggests that during wetter periods in the Quaternary, the lake likely drained westward towards the Büyük Menderes River (Erinç, 1967).

In the immediate vicinity of the lake, the land becomes bare as a result of the disappearance of vegetation cover due to reasons such as overgrazing and drought, the difficulty in renewing itself and the long time it takes to develop. While the mountains and slopes are generally covered with forests, the vegetation cover on the basin is primarily nonarborescent today.

Nonarborescent plants consists of thyme (*Thymus serpyllum*), snow thistle (*Acantholimon echinus*), meadowsweet (*Astragalus*), milk thistle (*Silybum marianum*), euphorbia (*Euphorbia*), cattail (*Verbascum*), reed (*Phragmites australis*). The main species belonging to the grass formation, which is widespread in the bottom and slope lands, are vetiver (*Peganum harmala*) and sea cowpea (*Salicornia europaea*) (Özdemir and Bahadır, 2010).

Arborescent plants consists of red pine (*Pinus brutia*), larch (*Pinus nigra*), juniper (*Juniperus*) and oak (*Quercus*) communities. Acorn oak (*Quercus ithaburensis*), a broad-leaved tree, is also observed in the forest flora. This situation reveals that the amount of precipitation in the mountainous area is quite high compared to the basin floor (Özdemir and Bahadır, 2010).

3. Materials and methods

3.1. Drilling and sampling

In 2009, the Alkim Alkali Kimya Company drilled a 602-m-deep borehole (Acıgöl2009-B03) beneath the lake floor in the dried-up area (37.866°N, 29.924°E, Fig. 1) of the Acıgöl basin (Helvacı et al., 2013; Akçer-Ön et al., 2017). 296 m sedimentary succession was recovered from the 341 m long drilling depth with an 87% core recovery rate. In the present study, the focus was directed to the upper 128 m of the retrieved sediment section. All samples and continuous U-channels are stored in a cold room (4 °C) in the Geological Engineering Department of Muğla Sıtkı Koçman University.

3.2. Magnetic susceptibility and density measurements

Gamma density measurements of Acıgöl2009-B03 sedimentary succession were determined a Geotek Multi-Sensor Core Logger (Weaver and Schultheiss, 1990; Weaver and Schultheiss, 1990) is scanned of 0.35 mm cm resolution at İTÜ-EMCOL Geochemistry Laboratory. MS and gamma density measurements were subjected to a first-order (Savitzky and Golay, 1964) filter through a window of five data points over the time axis and 17 data points over the depth axis. Afterwards, starting from the first sample, the data were resampled from every five and 17 samples, respectively.

3.3. Palynological analysis

A total of 123 samples from the upper 128 m of the Acıgöl2009-B03 sedimentary section were analyzed for palynomorphs for this research. Each sediment sample, weighing 20 g dry, underwent treatment with an acid mixture of HCl (35%) and HF (40%) to remove carbonate and silicate fractions. Palynomorph separation from the sediment was

achieved using a ZnCl₂ solution (density > 2). On each slide, at least 200 pollen grains (except Pinus) were counted using immersion oil under a transmitted light microscope. As an anemophilous (wind-pollinated) plant, Pinus produces large quantities of pollen grains, often overshadowing other pollen taxa in palynological assemblages. The overrepresentation of Pinus pollen complicates the interpretation of fossil pollen records, making it challenging to reconstruct past vegetation dynamics and climate change. For this reason, Pinus pollen was not counted and was not included in the analyses.

In this study woody plant data is used to construct the chronology of the Acıgöl2009-B03 sedimentary section. The pollen-based paleoenvironmental reconstruction will be the focus of an upcoming publication.

3.4. Radiocarbon and U-series dating methods

The ostracod shells of freeze-dried samples from three different depths where the suitable and abundant ostracod shells were (5.84 m, 6.58 m, 9.56 m) used for radiocarbon dating. The samples were washed through a 63 µm sieve. In situ (unbroken and unaltered) ostracod shells were collected under a stereomicroscope and cleaned via ultrasonication. Radiocarbon datings of the ostracod samples were performed at the Beta Analytic Laboratory in Florida, USA. Here, we report the calibrated samples (Table S1), which were processed using the IntCal20 calibration curve (Reimer et al., 2020) through the online age calibration program CALIB (Stuiver and Reimer, 1993). For U-series dating of sediments, ultrasonicated ostracod shells and gypsum crystals were used at GEOTOP research centre of the University of Quebec in Montreal. However, dating of ostracod shells were unsuccessful due to their low uranium concentrations ($\sim 197.4 \pm 0.9$ ppb with $^{234}\text{U}/^{238}\text{U}$ activity ratio of 1.322 ± 0.025) and high ^{232}Th content (417.278 ± 3.333 ppb), with latter indicating presence of detritus fraction despite the careful cleaning of the shells.

The method of Gázquez et al. (2020) was used for the chemical extraction of gypsum samples for the U-series analysis. Carefully cleaned gypsum samples were powdered and dissolved in hot dilute HCl (~ 0.5 M) by magnetic stirring. After total dissolution, ~ 7 mg of FeCl₃ carrier was added and left overnight. U and Th were separated from the bulk material via Fe(OH)₃ precipitation by adding dry NH₄(OH) until the pH is approximately ~ 8 – 9 . The precipitate was recovered by centrifuging, washed with MilliQ water then dissolved in HCl 6 M.

For U-Th separation in gypsum sample solutions, the method of Edwards et al. (1986) was conducted using an AG1X8 anionic resin bed. The Th and U-Fe fractions were retrieved by elution with 6 N HCl and H₂O, respectively. The purification of the U fraction was done using 0.2 ml U-Teva (Eichrom® Industries) resin volume. The U-Fe separation was performed by elution using 3 N HNO₃ (Fe fraction) and 0.02 N HNO₃ (U fraction). The purification of Th was performed by elution using 2 ml AG1X8 resin in 7 N HNO₃ and elution with 6 N HCl. Final purification of Th was carried out on a 0.2 ml AG1X8 resin in 7 N HNO₃, and Th was eluted with 6 N HCl. The U and Th fractions were evaporated to dryness and then dissolved with 73 ml of HNO₃ (2%). The U and Th fractions solutions were analyzed using a multi-collectors inductively coupled plasma mass spectrometry (MC-ICP-MS) Nu instrument using ions counter (IC0). Mass bias corrections were performed using the $^{236}\text{U}/^{233}\text{U}$ certified ratio of the spike.

As estimated from replicate measurements of standards, the overall analytical reproducibility is usually better than 0.5% for U concentrations and $^{234}\text{U}/^{238}\text{U}$ ratios and ranges from 0.5% to 1% for $^{230}\text{Th}/^{234}\text{U}$ ratios ($\pm 2 \sigma$ error range).

3.5. Tephrochronological methods

Sample material of the visible tephra layer at approximately 47.6 m depth in the Acıgöl sequence was embedded in epoxy resin, sectioned, and polished for electron probe microanalyses. The major element compositions of the individual glass shards were obtained at the

ACIGÖL2009-B03

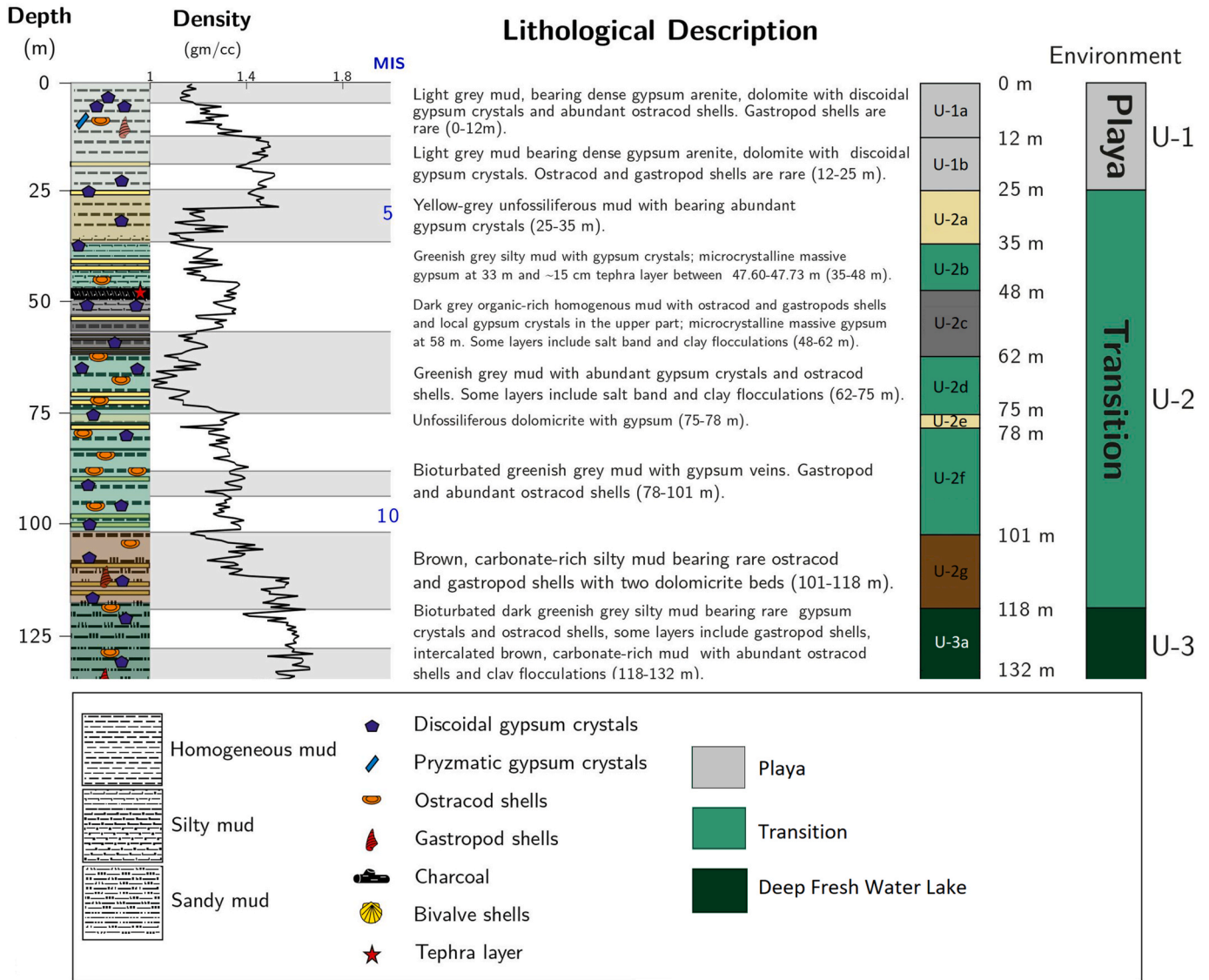


Fig. 3. Lithostratigraphy of the Acigöl2009-B03 sediment core, with gamma density and ostracod and evaporite mineral abundance shown along the stratigraphic column.

Helmholtz Centre Potsdam-GFZ German Research Centre for Geosciences using a JEOL JXA-8230 instrument with an accelerating voltage of 15 kV, a current intensity of 10 nA and a beam diameter of 10 μ m. Secondary glass standards used the Lipari obsidian. The Lipari obsidian was used as secondary glass standard (Hunt and Hill, 1996; Kuehn et al., 2011). For comparability, the analytical data were normalized to 100 wt % on an anhydrous and volatile-free basis.

3.6. Dynamic time warping

The DTW algorithm was employed to develop the age-depth model. Widely utilized for comparing time series, DTW utilizes dynamic programming techniques. It has been extensively applied in previous studies with stratigraphic and paleoclimatic investigations of geological time series (Haam and Huybers, 2010; Hay et al., 2019; Ajayi et al., 2020; Hagen et al., 2020; Baviile et al., 2022).

In DTW, two vectors, denoted as n and m , are compared by constructing a distance matrix. In this matrix, each (i, j) element defines the Euclidean distance between the i th element of vector n and the j th

element of vector m . In DTW, one of these vectors serves as the target or reference data, while the other vector is considered the query or candidate data. The objective is to find the optimal alignment path that minimizes the cumulative distance between corresponding elements of the target/reference data and the query/candidate data (Cassisi et al., 2012). While finding the optimal alignment, DTW sets some constraints to limit the paths, which are monotonicity, continuity, boundary conditions, warping window, warping function, slope constrain. By applying these constraints, Dynamic Time Warping (DTW) identifies the optimal alignment between the two vectors through iterative processes utilizing dynamic programming methods. This technique allows for flexible matching of time series that may vary in speed or temporal distortions, enabling effective comparison and alignment in various applications, including paleoclimatic and stratigraphic studies. In our study, the arboreal pollen percentage of core ACIGÖL2009-B03 was designated as the “query” or “candidate” set.

3.7. QAnalySeries

QAnalySeries is a versatile, cross-platform tool designed specifically for time series tuning and analysis (Kotov and Paelike, 2018).

3.8. Bayesian age-depth modeling

We used the Bchron package (Haslett and Parnell, 2008) in R for constructing the final age-depth model. This package offers a Bayesian solution through a slightly modified Markov chain Monte Carlo sampling method for the piecewise linear regression model. In this study, we ran 10,000 iterations, discarding the first 2000 as burn-in to ensure convergence.

4. Results and discussion

4.1. Lithological description

The 128 m long Acıgöl2009-B03 sedimentary sequence is divided into three main lithological units and 10 subunits based on lithological properties (Fig. 3). The uppermost Unit 1 (U-1), between 0 and 25m interval, includes subunits U-1a and U-1b, which is light gray mud with abundant gypsum, dolomite and other sulfate minerals such as mirabalite ($\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$) and lesser amounts of celestite (SrSO_4). This unit also includes ostracods and rare gastropod shells. Gypsum in U-1 exists as gypsum arenite and as discoidal and prismatic crystals. The gamma density variations are inversely proportional to porosity in the sedimentary sequence. These variations depend on the density of coarse-grained materials (typically crystals, mollusks etc). It is observed that density decreases from the bottom to the top of the core. In the playa environment, the density of macrocrystals presumably causes device readings to reflect changes in crystal size rather than actual density. Therefore, density interpretations should be excluded from analyses in the playa environment.

The middle Unit 2 (U-2) (25–118-m interval) includes subunits U2-a to U2-g and is composed of light to dark gray mud, which contains abundant gypsum and ostracod shells and rarer gastropod and bivalve shells (Fig. 3). Other sulfate minerals include mirabalite and lesser amounts of celestite at some depths (e.g., at 26, 62, 74, 110 and 124 m). Carbonate minerals in this unit include calcite, dolomite and aragonite. Subunits U-2f and U-2g include two unfossiliferous dolomicrite beds and abundant gypsum. U-2b contains a 15 cm thick tephra layer between 47.60 and 47.75 m interval (see Sec. 2.4-Tephrochronology). U-2g shows an upward decreasing intermediate gamma density values (from bottom to top), with relatively low gamma density (on average 1.2 g/cm³) for subunits U-2d, U-2b and U-2a. The low density values of these three units are the result of the high porosity between the coarse evaporite mineral grains.

Unit 3 (U-3) includes subunit U-3a and consists mainly of greenish gray to dark gray homogeneous mud with some intercalations of brown carbonate-rich mud, local peat, coalified wood, and shows bioturbation. U-3 is relatively rich in fossils (gastropods and ostracods) and organic matter, and is mainly devoid of evaporate minerals.

According to the lithological description of the Acıgöl core, three different environments have been identified: a deep lake environment, a transitional environment, and a shallow, hypersaline playa environment. From 118 m to 25 m, a transition from the deep lake environment is observed. From 25 m onward, the core indicates the presence of today's playa lake conditions. Accordingly, between 485 and 420 kyrs ago, deeper conditions were observed, followed by a gradual transition between 420 and 75 kyrs ago, and from 75 kyrs to the present, a shallow hypersaline environment is evident. We infer that the system has transformed from a deep lake to a playa lake as it shallowed, evidenced by crystal formations that increase in size and grow progressively from the bottom to the top. These three main periods include recurring changes in lake water level and salinity, represented by the lithological

subunits, which suggest shallowing intervals even during the freshwater period as indicated by some peat intervals in U-3, and evaporitic intervals in U-2 and U-1.

4.2. Radiocarbon dating

In lakes, the calibration of radiocarbon dates depends on dissolved inorganic carbon containing radiocarbon-dead carbon (Philippsen, 2013). The magnitude of the required correction can reach several thousand years, particularly for alkaline, highly saline lakes or those fed by rivers flowing over carbonate-rich terrains (An et al., 2018; Makaroglu et al., 2016; Stein et al., 2004). Furthermore, secondary recrystallization processes in ostracod shells have the potential to significantly alter the ages of samples obtained from radiocarbon dating (An et al., 2018; Lai et al., 2014; Liu et al., 2024; Long et al., 2022; Petchey et al., 2022; Song et al., 2015; Zamanian et al., 2016). However, neither the proportion of radiocarbon-dead carbon in ostracod shells nor the impact of recrystallization processes is known for the radiocarbon dates from Lake Acıgöl. For this reason, the C-14 datings have been provided without taking into account the effects of the hard water effect or recrystallization processes. Even if such information were available, the resolution of our age-depth model, which relies primarily on the dating points for validating the final model, suggests that the results would not change substantially.

4.3. Tephrochronology

The 15 cm thick tephra layer at 47.6 m depth is characterized by highly vesicular glass shards that show a homogeneous high-silica rhyolitic composition. Raw (non-normalized) major element glass compositions yield contents in SiO_2 at 73.7–74.7 wt%, Al_2O_3 at 11.7–12.2 wt%, low concentrations in FeO (0.4–0.6 wt%) and CaO (0.5–0.6 wt%) and alkali ratios $\text{K}_2\text{O}/\text{Na}_2\text{O}$ of 1.1–1.4 (Table S4). The composition of tephra layer matches best with that of the Kos Plateau Tuff (KPT) (Fig. S2). The KPT was derived from a large, caldera-forming eruption centered south of the Island of Kos in the southeastern Aegean Sea (e.g., Allen and Cas, 1998), and is dated by $^{40}\text{Ar}/^{39}\text{Ar}$ at 161.3 ± 1.1 ka (Smith et al., 1996). The KPT occurs as the marine W-3 tephra in deep-sea sediments in the southern Aegean Sea region east of Crete (e.g., Federman and Carey, 1980; Vinci, 1985; Satow et al., 2015). In the Acıgöl sequence, it forms a valuable time marker within the penultimate glacial period and is a reliable isochron for use in our age-depth modeling and for comparison with marine paleoclimate records.

4.4. U-series chronology

Gypsum samples from 35.9 m, 62 m and 116.5 m core depths were dated (Table S5). The results show that while the sediments have a relatively high uranium content of up to 1 ppm, the uranium content in the gypsum samples is very low (3–18 ppb). All samples indicate contamination by detrital fractions as highlighted by the ^{232}Th content. Only three samples at 35.9m provided acceptable uncorrected ^{230}Th – ^{234}U – ^{238}U ages of 125.3, 130.5 and 141.2 ka. These uncorrected ages should be considered as the maximum ages due to detrital contribution and must be corrected by subtracting the ^{230}Th linked to ^{232}Th from the measured ^{230}Th . To perform this correction, we used the average crustal model suggested by Ludwig and Paces (2002). Specifically, we use $^{232}\text{Th}/^{238}\text{U} = 1.21 \pm 0.65$, $^{230}\text{Th}/^{238}\text{U} = 1 \pm 0.1$, $^{234}\text{U}/^{238}\text{U} = 1 \pm 0.1$ as isotopic activity ratios of the detrital component. The results are listed in Table S3 as corrected ages. Despite the large uncertainties, samples from this depth can be assigned to MIS 5. It is worth mentioning that even with the uncorrected age or using other approaches for the correction method (Kaufman, 1993; Develle et al., 2011), the samples from this depth agree with the attribution to MIS 5. The corrected ages from this depth indicate a relatively constant initial uranium isotopic activity ratio value, with an average value of

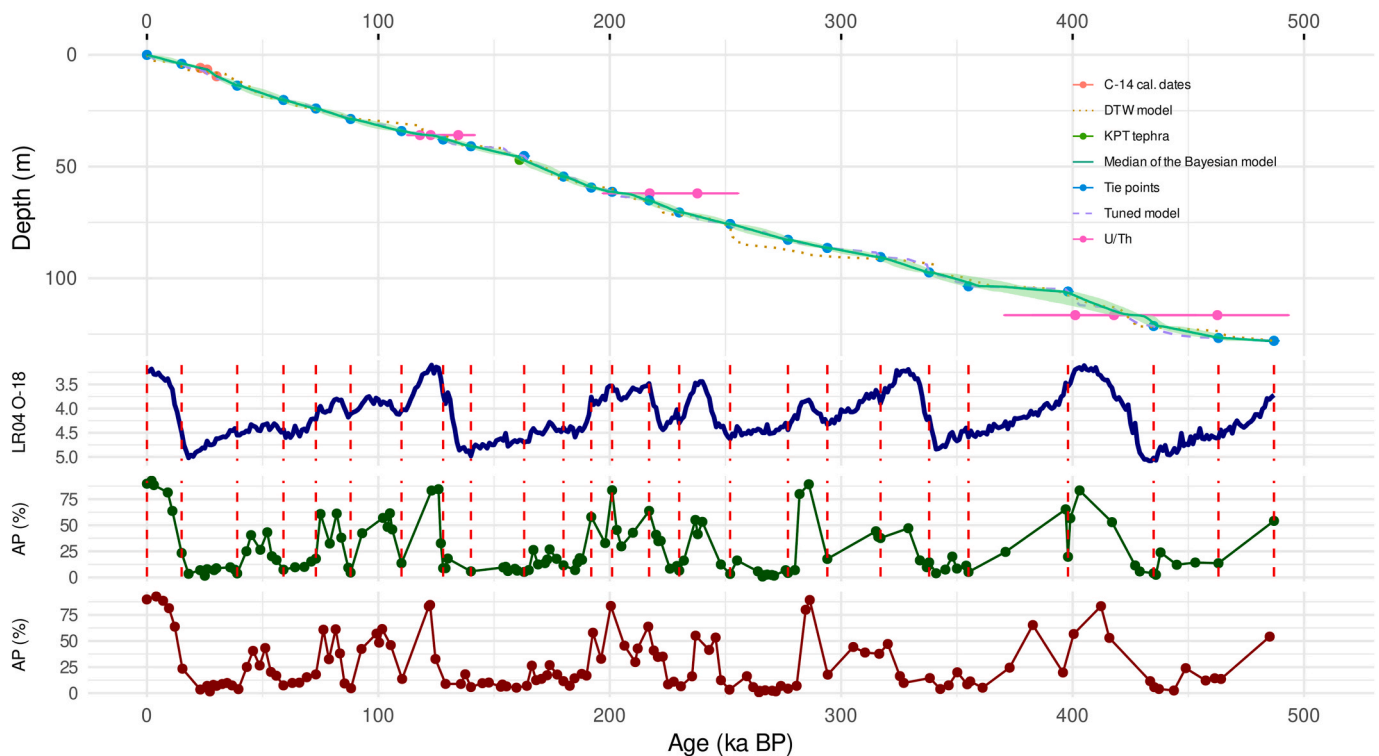


Fig. 4. The top figure displays the tie points, control points, the dynamically time-warped alignment (dashed red line), the tuned age-depth model (smooth blue line), and the final Bayesian age-depth model with 89% credible intervals for the Acigöl2009-B03 core. The lower panel presents 25 tie points assigned between the LR04 global benthic isotope stack (Lisiecki and Raymo, 2005) and the arboreal pollen percentage. The bottom plot shows the final age model obtained through the Bayesian process.

$$(^{234}\text{U}/^{238}\text{U})_0 = 1.471 \pm 0.017.$$

Samples from core depths of 62 m and 116.5 m have higher detrital fractions, as indicated by the higher ^{232}Th content. The average crustal model is also used for the correction, but such a correction would result in large errors or undetermined error bars in the calculated corrected ages. For these two samples, we performed U/ U_0 age calculations in a manner similar to that used by Ghaleb et al. (2019) (Table S5). The advantage of this approach is the possibility to go further back in time with reasonable errors in the calculated corrected ages. These calculated U/ U_0 ages allowed us to attribute the depths of 62 m and 116.5 m to MIS 7 and MIS 11, respectively.

4.5. Age-depth model

Our primary assumption regarding the query dataset is that the 128th meter of the core corresponds approximately to 490 ka BP. This estimation is based on U/Th dating conducted at the 116th meter of the core (Tables S3 and S5), combined with the observation of arboreal pollen peaks observed up to the 116-m depth.

The LR04 $\delta^{18}\text{O}$ data provided relatively high-resolution information with 491 data points for the time interval 0 to 490 ka BP, whereas the Acigöl2009-B03 pollen data comprised 123 data points. To align these datasets effectively, the length of reference data was addressed by resampling. First, the reference data was smoothed using a linear Savitzky-Golay filter with a window length of three (Savitzky and Golay, 1964). Then, the data was resampled to one-third resolution, resulting in the restructuring the dataset from 491 original data points to 164 data points. This process allowed for a more reasonable and aligned comparison between the high-resolution marine oxygen isotope record and the arboreal pollen percentage data from core Acigöl2009-B03.

In the context of DTW, the alignment of query data (x-axis) with reference data (y-axis) reveals the optimal warping path through a cost matrix. Ideally, a perfect match between the datasets would result in a

linear diagonal alignment. However, due to temporal distortions or variations in the data, the DTW algorithm calculates the optimal path that minimizes the cumulative distance between corresponding points of the two series. However, resampling has been important for the model as outlined by Ratanamahatana and Keogh (2004). The inset in Fig. 4 illustrates this optimal warping line, which is determined by the DTW algorithm. This line represents the path where the query data points are aligned with the reference data points in the most optimal manner possible, considering the constraints and distortions present in the datasets. This technique is crucial for comparing time series that may exhibit shifts, accelerations, or decelerations over time, as observed in paleoclimatic and stratigraphic studies (Hagen and Harper, 2023).

After standardization of both the arboreal pollen and LR04 benthic $\delta^{18}\text{O}$ stack records, the “dtw” function from the dtw package (Giorgino, 2009; version 1.23–1) in R software (R Core Team, 2024) was applied to perform the alignment. The DTW algorithm computed the optimal warping path between the datasets, providing alignments as output in a data frame format. These alignments effectively served as the age-depth model, determining which depths in the query (pollen) data corresponded to which ages in the reference (LR04 $\delta^{18}\text{O}$) data ((Giorgino, 2009). This approach facilitated the construction of a chronological framework for the Acigöl2009-B03 sedimentary section, integrating both pollen and marine isotope data.

Due to limitations stemming from varying resolutions and the use of different proxy data, coupled with inherent constraints of the DTW algorithm (see section 3.7) minor discrepancies emerged in the relationship between the reference data (LR04 benthic $\delta^{18}\text{O}$ stack) and the query data (arboreal pollen percentage of core Acigöl2009-B03) within the age-depth model. To address these discrepancies, the age-depth model of the Acigöl2009-B03 sedimentary section was carefully refined using the QAnalyzeSeries software. In this reconstruction, 25 tie points were selected to match the arboreal pollen percentage data from core Acigöl2009-B03 with the LR04 benthic $\delta^{18}\text{O}$ stack marine oxygen

isotope record against age (Fig. 4 and Table S2). The tie points were primarily chosen by correlating the local maxima and minima of the dynamically time-warped arboreal pollen time series with the local minima and maxima, respectively, of the LR04 stack isotope record (Fig. 4). This approach aimed to establish a more accurate and consistent age-depth relationship between the two datasets, thereby refining the chronological framework of the sedimentary section from core Acıgöl2009-B03. Furthermore, three radiocarbon dates, three U/Th dates, tephrochronology were used to verify the age-depth model (Table S3).

Upon constructing an age-depth model using tie points, we applied a Bayesian statistical approach to incorporate the dated points and estimate the uncertainty of the age-depth model. This was achieved through the Bchron package (Haslett and Parnell, 2008) in R, which employs a monotone stochastic process to determine piecewise linear paths between depths with known ages. As input, we utilized U-Th ages, tephra layers, and uncalibrated radiocarbon ages, along with their respective uncertainties. Additionally, the 25 tie points were included as dating points, each assigned an uncertainty of 2000 years.

The age-depth model constructed for the Acıgöl2009-B03 sedimentary section reveals that the upper 128 m of the section spans the last 485 ka (Fig. 4 and Fig. S3). This time interval corresponds approximately to MIS 1 through 12 and nearly half of MIS 13. Our model, with 25 tie points and 10 age control points, provides a great improvement with a resolution of approximately 20 ka (Fig. S4) over the previously constructed age-depth model of Demory et al. (2020), which is based on linear interpolation between only three paleomagnetic tie points for the same 296 m long chronostratigraphic interval (Fig. S5).

4.6. Environmental and vegetation variability over the last 485 ka

Based on the lithological description of the Acıgöl core, three environments have been identified: a deep lake environment down to 118 m depth, a transitional environment between 118 and 25 m depth, and a shallow, hypersaline playa environment from 25 m depth to the core top. According to the age-depth model, deeper conditions existed from 485,000 to 420,000 years ago, followed by a gradual transition from 420,000 to 75,000 years ago. From 75,000 years to the present, a shallow, hypersaline environment is evident. The transition from a deep lake to a playa lake is marked by a noticeable increase in evaporite crystals, which grow progressively larger from bottom to top, along with a decrease in gastropod fossils.

The mud and mudstone layers, formed in a deeper water environment, reveal two main stages of change. Around 420,000 years ago, dolomitic mudstones developed in sequences of carbonate-gypsum and gypsum-carbonate alternations. From approximately 75,000 years ago to the present, the lake deposits show a shallowing trend with playa characteristics and an enrichment in Na-sulfate. This shallowing during both periods could result from climatic or tectonic influences, though this study cannot determine the extent of tectonic impact. The onset of the changes of the character of the lake appear to have begun during MIS 11 and MIS 4, suggesting that shifts following these temperate (interglacial) and cold (glacial) periods may have altered the lake's structure. Increased aridity, along with tectonic adjustments in water channels or basin uplift, may have expanded the lake surface, lowering lake levels and raising evaporation. Presently, Acıgöl is primarily fed by groundwater, and aridification or faulting that disrupts groundwater flow may have gradually reduced the lake's water budget, causing shallowing and eventual transition to playa characteristics. As revealed by Helvacı et al. (2013), the regional expansion stage of the lake corresponds to early Pliocene. Therefore, gradual shallowing of the lake may be due to a tectonic influence causing rivers feeding the basin to change course, but this requires further research.

Endorheic lakes like Acıgöl are highly sensitive to climatic fluctuations and depositional hiatuses could exist in endorheic lake deposits (see Ding et al., 2020, 2021, 2024; Fu et al., 2017; Zhang and Liu, 2012).

In hyper-arid basins as Qaidam Basin in the Tibetan Plateau, water bodies often exhibit extreme salinity and limited vegetation cover, leading to lower arboreal pollen deposition. In contrast, Lake Acıgöl's semi-arid climate supports more substantial vegetation in its catchment area, leading to distinct pollen deposition patterns in their respective sediment records. To ensure that the sedimentary processes in the Lake Acıgöl core are continuous and free of major hiatuses in the presented interval, sediment cores from lakes in similar climatic settings were compared to identify regionally consistent patterns and potential disruptions. Arboreal pollen percentage data of Lake Acıgöl correlates well with pollen data from other Eastern Mediterranean lakes, including those from Tenaghi Philippon (Tzedakis et al., 2006), Lake Ohrid (Wagner et al., 2019) and Lake Van (Litt et al., 2014). The period corresponding to MIS1 (present day to approximately 15 ka, corresponding to 4 m sediment depth) displays an increasing AP pattern. This suggests that this period, was warm and humid enough to support Mediterranean type arboreal plants in the region. The following depth between 4 and 25 m reveals the transitions between MIS 2, 3 and 4. The depths between 25 m and 40 m, 40 m–59.5 m and 59.5 m–75.7 m show distinct glacial/interglacial transitions with 75 ka, 132 ka and 252 ka, respectively. In these zones, interglacial conditions are evidenced by increased AP. These depths are also well correlated with MIS 5, 6 and 7. If it is examined the ages from 329 ka to 282 ka, it can be observed a decreasing trend in pollen percentage. However, there is a serious increasing pollen percentage between 282 ka (84,9 m) and 294 ka (86, 42 m). This phenomenon has been documented in various studies (Follieri et al., 1998; Tzedakis et al., 2002; Fletcher et al., 2013) in southeastern Europe and indicates the sensitivity of vegetation to regional climate changes. This is also observed in lake Ohrid and Tenaghi Philippon results. The depths from 282 ka (84,29 m) to 485 ka (128 m) shows clear glacial/interglacial behaviour same as the other study areas as Lake Van, Lake Ohrid and Tenaghi Philippon.

The last 485,000 years of sediment records from Acıgöl show that arboreal plants generally decrease during cold/dry periods (such as MIS 2, 4, 6, etc.) and increase during temperate/wet periods (such as MIS 1, 3, 5, etc.). Local authigenic factors also play a crucial role in shaping the arboreal pollen record. Approximately 25% of the lake's volume evaporates during transitions between dry and pluvial periods, resulting in a reduction of the lake area by 55–61%. These hydrological changes can influence local vegetation distribution, thereby affecting the types and quantities of arboreal pollen deposited in the lake sediments (Karaman, 2017). Additionally, the lake's hypersaline and alkaline conditions, can influence the preservation and composition of pollen assemblages. Authigenic carbonate precipitation, driven by both abiotic and microbial factors, may affect the sedimentation processes and the incorporation of pollen into the lake's sediment record (Balci et al., 2018). However, comparative results from the Acıgöl2009-B03 sediment core indicate that any differences arising from the limnological characteristics of Lake Acıgöl or the regional setting are negligible in the context of this study. The reduction of arboreal plants in relatively dry/cold periods, largely due to decreased precipitation, plays an important role in identifying glacial and interglacial periods of the Pleistocene (e.g., Tzedakis et al., 2006; Wagner et al., 2019). Similarly, the shift from woodland to shrubland during these cold/dry periods increases physical erosion and, as observed in Acıgöl, enhances the influx of clastic material into the basin through wind transport. The rise in magnetic susceptibility (MS) values, representing clastic influx, during the cold/dry last glacial cycle is likely due to the reduction of surrounding vegetation and increased erosion, as also suggested by Demory et al. (2020) and Francke et al. (2016) for Lake Ohrid.

However, an alternative hypothesis suggests that the eastern Mediterranean was at least as humid during glacial periods as it was in the Holocene (Sarıkaya et al., 2008, 2014; Torfstein et al., 2013; Candaş et al., 2020; Köse et al., 2022). Another explanation for the high detrital inputs during glacial periods could be that cold and relatively dry conditions led to low vegetation density, combined with an increase in

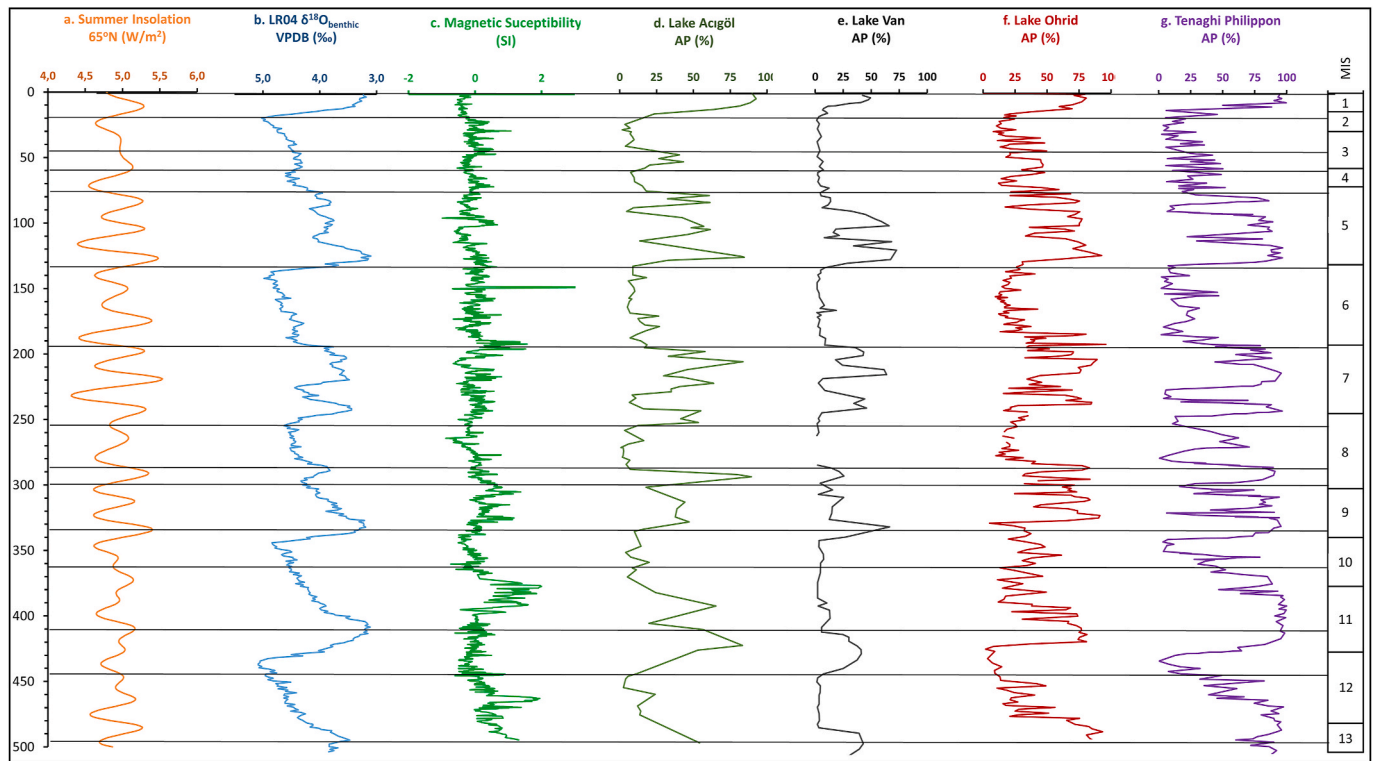


Fig. 5. Correlation of Acıgöl2009-B03 with other global and regional data for the last 485 ka. a. Summer insolation curve of 65°N (Berger and Loutre, 1991); b. The global benthic isotope stack LR04 (Lisiecki and Raymo, 2005); c. Magnetic susceptibility results of Acıgöl2009-B03; d. Arboreal pollen percentage from Lake Van (Litt et al., 2014); e. Arboreal pollen percentage from Lake Van (Litt et al., 2014); f. Arboreal pollen minus pinus pollen from Lake Ohrid (Wagner et al., 2019); g. Arboreal pollen percentage from Tenaghi Philippon (Tzedakis et al., 2006).

spring-summer precipitation compared to the current interglacial winter-dominated precipitation pattern (see Fig. 5 in Davis et al., 2024). The present-day climate of the Acıgöl region is characterized by cold, rainy/snowy winters and hot, relatively dry summers (Fig. S1). During glacial periods, this region may have experienced greater spring or summer humidity due to the southward displacement of the extra-tropical low-pressure and subtropical high-pressure systems, driven by the strengthening of the polar anticyclone (Harrison and Digerfeldt, 1993; Ön and Özeren, 2019). With additional data from Acıgöl, future studies will provide further clarification on the increase in clastic input during glacial periods.

5. Conclusions

In this study, we present a refined chronology of the upper 128 m of a 296-m-long core section (Acıgöl2009-B03) retrieved from Lake Acıgöl, located in southwestern Anatolia. The chronology was established by aligning the pollen percentage of arboreal plants with the global benthic stack marine oxygen isotope curve using DTW, followed by fine-tuning and consequently Bayesian modeling. The reliability of this chronology is verified by radiocarbon and U/Th datings, and tephra ages.

The constructed age-depth model reveals that the 128-m section of the Acıgöl2009-B03 sequence spans approximately the last 485,000 years. Specifically, it covers MIS 1 to MIS 12 and approximately half of MIS 13. The sedimentary records of the past 485,000 years from Acıgöl reveal a general decline in arboreal plant prevalence during cold and dry periods, contrasted with an increase during temperate and wet periods. Lithological analysis of the Acıgöl core indicates the presence of three distinct depositional environments: a deep lake environment, a transitional environment, and a shallow, hypersaline playa environment.

This model provides a robust framework for future studies aimed at understanding the sedimentary history and paleoenvironmental changes in the Acıgöl region over Quaternary timescales.

CRediT authorship contribution statement

Hüseyin Çaldırak: Writing – original draft, Writing – review & editing, Visualization, Validation, Formal analysis, Software, Methodology, Conceptualization. **Z. Bora Ön:** Writing – original draft, Writing – review & editing, Visualization, Supervision, Investigation, Validation, Formal analysis, Software, Methodology, Conceptualization. **M. Serkan Akkiraz:** Investigation, Resources, Data curation, Writing – review & editing. **M. Namık Çağatay:** Writing – review & editing. **Bassam Ghaleb:** Investigation, Resources, Data curation, Writing – review & editing, Writing – original draft. **Sabine Wulf:** Investigation, Resources, Data Curation, Writing – review & editing, Writing – original draft. **K. Kadir Eriş:** Investigation, Resources, Data Curation, Writing – review & editing, Project administration. **Dursun Acar:** Investigation, Resources, Data Curation, Writing – review & editing. **Jérôme Kaiser:** Writing – review & editing. **Sena Akçer-Ön:** Conceptualization, Methodology, Validation, Investigation, Resources, Data Curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.quaint.2025.109710>.

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